ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** magnetic field	Definition.
2	**(d)** a magnetic field	Oersted's discovery.
3	**(b)** Fleming's left-hand rule	Force direction.
4	**(d)** tesla	T.
5	**(d)** an electric current	Current deflects the needle.
6	**(a)** the field	Forefinger: field.
7	**(d)** repel each other	N-N repel.
8	**(d)** in the opposite direction	The field direction reverses.
9	**(c)** force or motion	Thumb: motion.
10	**(a)** attract each other	N-S attract.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The region around a magnet where its force is felt (2).	2
12	Thumb along the current (1); Fingers curl in the direction of the field (1).	1+1
13	Michael Faraday (2).	2
14	North pole (1); South pole (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Closed curves (1); Outside: N to S (1); Inside: S to N (1).	1+1+1
16	Forefinger: field (1); Middle finger: current (1); Thumb: force (1).	1+1+1
17	Live: red (1); Neutral: black (1); Earth: green (1).	1+1+1
	OR: They are closed curves (1); They do not cross (1); Closer lines mean a stronger field (1).	1+1+1
18	Thumb: direction of motion (1); Forefinger: field (1); Middle finger: induced current (1).	1+1+1
	OR: It protects the appliance from excess current (1); It melts when the current is large (1); Prevents fire hazards (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	They emerge from the north pole and end at the south pole (1); They form closed curves (1); The tangent gives the direction of the field (1); Closeness shows strength (1); No two lines intersect (1).	1+1+1+1+1
	OR: Principle: electromagnetic induction (1); Armature coil, magnets, slip rings, brushes (1); Coil rotates in the field (1); Current is induced and changes direction every half turn (1); Output is alternating current (1).	1+1+1+1+1
20	A straight wire is connected to a battery and a key (1); A compass is placed near the wire (1); On passing the current the needle is deflected (1); Reverse the current: deflection reverses (1); Current produces a magnetic field (1).	1+1+1+1+1
	OR: DC has a fixed direction, AC reverses (1); DC source: cell (1); AC source: generator, mains (1); AC can be transmitted over long distances (1); With less loss of energy (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** concentric circles	Circular.
2	**(c)** Faraday	Michael Faraday.
3	**(a)** north-south	Because of the earth's field.
4	**(a)** right-hand thumb rule	Thumb along the current.
5	**(d)** Generator	Converts motion to electricity.
6	**(d)** compass	A small magnet.
7	**(d)** a solenoid	Cylindrical coil.
8	**(b)** induced current	Induced by a changing field.
9	**(a)** magnetic field lines	They get magnetised.
10	**(a)** uniform	Parallel lines.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A coil of many turns of insulated wire (1); Of a cylindrical shape (1).	1+1
12	To convert mechanical energy into electrical energy (2).	2
13	From the north pole to the south pole (2).	2
14	A magnet formed by a current in a coil around a soft iron core (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place the magnet below a sheet of paper (1); Sprinkle iron filings (1); Tap the paper; filings align along field lines (1).	1+1+1
16	Electromagnetic induction (1); A coil rotating in a field (1); Induces a current (1).	1+1+1
17	220 V (1); Each gets the same voltage (1); Each works independently (1).	1+1+1
	OR: A wire is placed near a compass (1); The current is passed (1); The needle is deflected (1).	1+1+1
18	AC changes direction (1); DC flows in one direction (1); AC from mains and DC from cells (1).	1+1+1
	OR: Fuse (1); Earth wire (1); MCB (miniature circuit breaker) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Concentric circles around the wire (1); Direction by the right-hand thumb rule (1); Field increases with current (1); Field decreases as the distance increases (1); The field is stronger near the wire (1).	1+1+1+1+1
	OR: Mains supply of 220 V enters through the main fuse (1); Live (red) and neutral (black) wires (1); Earth (green) wire connected to the appliance body (1); Appliances in parallel (1); Fuse in live wire (1).	1+1+1+1+1
20	Solenoid is a coil of many circular turns (1); Field inside is uniform (1); Outside it is like a bar magnet (1); Soft iron rod is placed inside (1); It becomes an electromagnet when current flows (1).	1+1+1+1+1
	OR: Short circuit: live and neutral wires touch (1); Very large current flows (1); Overloading: too many appliances in one socket (1); Use fuse or MCB (1); Do not overload sockets (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** alternating current	Direction changes.
2	**(c)** Iron	Ferromagnetic.
3	**(d)** soft iron	Easily magnetised and demagnetised.
4	**(b)** direct current	One direction.
5	**(a)** iron nails	They contain iron.
6	**(b)** temporary magnet	Works only with current.
7	**(a)** DC	A cell gives a current in one direction.
8	**(b)** north pole	N to S.
9	**(b)** electric bells	Common use.
10	**(c)** 220 V, 50 Hz AC	Standard mains.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Current that changes its direction periodically (2).	2
12	Hans Christian Oersted (1); In 1820 (1).	1+1
13	Fleming's left-hand rule (2).	2
14	Safety against electric shocks (1); Leakage current goes to earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Permanent magnet cannot be switched off (1); Electromagnet is temporary (1); Electromagnet strength can be changed (1).	1+1+1
16	50 Hz (1); 50 complete cycles per second (1); Direction changes 100 times per second (1).	1+1+1
17	Slip rings rotate with the coil (1); Brushes press on the rings (1); They carry the current to the external circuit (1).	1+1+1
	OR: Current (1); Number of turns (1); Material of the core (1).	1+1+1
18	Easy to transmit over long distances (1); Less power is wasted in the lines (1); Generated easily in power stations (1).	1+1+1
	OR: Current produced due to a changing magnetic field (1); Seen in a closed coil (1); Detected by a galvanometer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Forefinger: field (1); Middle finger: current (1); Thumb: force (1); Fingers are mutually perpendicular (1); Force depends on current, field strength and angle (1).	1+1+1+1+1
	OR: Metal body may become live due to a fault (1); A person touching it gets a shock (1); Earth wire gives a low-resistance path (1); The current goes to the ground (1); The shock is avoided (1).	1+1+1+1+1
20	A coil is connected to a galvanometer (1); A magnet is moved into the coil (1); The galvanometer deflects (1); On withdrawal it deflects the other way (1); There is no deflection when the magnet is at rest (1).	1+1+1+1+1
	OR: Generator: electromagnetic induction (1); Electromagnet: magnetic effect of current (1); Generator: mechanical to electrical (1); Electromagnet: electrical to magnetic (1); Both use coils (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND